

Summary Report

Bollinger Bands & Volume Divergence

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Introduction

Initially given the inclusion of stock pairs with similar industry and size, an analysis based on a pair-trading mean reversion strategy was prepared. However, following a lack of evidence of cointegration between pairs and therefore no potential for stationarity or mean-reversion of spreads between pairs, the trading approach was changed to focus on individual asset prices based on momentum and volume indicators. This was since the lack of robust indication of cointegration ruled out most asset correlations during this period as temporary or spurious and also negatively affected the potential for profitable strategies of other “soft correlation” or mean-reversion strategies including utilising a Kalman filter or correlation-based pairs.

Back-testing assumptions included an initial capital of \$100,000 and transaction fees of 0.01% for each position entered, with the simulated portfolio trading from June of 2020 to June of 2025. Furthermore, missing or erroneous data was handled to forward fill with datapoints from the previous trading day.

Strategy Description

The trading strategy involved generating bullish or bearish alpha signals when consecutive price movement over the course of three days approached within ten percent of a Bollinger band (rolling window of twenty days & two standard deviations) relative to the upper and lower bound range and was combined with a consecutively weakening volume.

The divergence in price movement and volume represented decreased momentum and support for trend-continuation. Additionally, the approach towards a Bollinger-band acts as a potential reversal indicator as traders entering in the last twenty-days, either are commonly exiting losing-trades at the average breakeven point or taking profit at the two standard deviation mark. These combined indicators act as a short-term reversal signal.

The exit strategy involved exiting the position five trading days after initial entry to maintain short-term exposure. Additionally, a stop loss was implemented at five percent drawdown from the initial entry price to take into account when price movements regain momentum into trend-confirmation.

Methodology

The dataset was formatted with the NumPy library with prices set as two decimal-point floats and volume set as integers, into a Pandas library data-frame sorted by date. For (1) the cointegration analysis, combinations of all pairs were handled through an Engle-Granger cointegration test that utilised a rolling window of one year over a training period of two years. Prices were also log-normalised to adjust dataset closer to a normal distribution. The pairs were then evaluated based on when they met a p-value of five percent and its equivalent test statistic over the total number of rolling windows.

Furthermore, the stability of the hedge ratio from the spread between pairs was evaluated based on its standard deviation. These criteria ensure a statistically significant rejection of the null hypothesis of no cointegration and a stable hedge ratio, which ensures stability of the

relationship between pairs. If these criteria were met, the spread could then be tested again for stationarity and to determine half-life with an Augmented Dickey-Fuller test. This could then be utilised along with Z scores for the spread to determine which pairs had reached unsustainable divergence in spreads and was expecting mean reversion in the short term as to ensure capital isn't being committed to pairs with an excessively long half-life.

Unfortunately, the cointegration tests resulted in an inconclusive rejection of the null hypothesis and the trading strategy was changed to utilise (2) momentum and technical indicators. Bollinger bands were built on close prices for each symbol over a twenty-day window with upper and lower bounds set at two standard deviations. Diverging price and volume signals were built on a three-day bullish or bearish consecutive price movement for close prices and a three-day consecutive decrease in volume. These datapoints were taken as a series from each individual asset's close prices and respective volume. These were then passed through the signal generation model to create a data-frame of long or short signals with the respective date and price.

Exit signals due to breach of five percent stop loss mechanism or five-day holding period were added to these data-frames. Additionally, if during the five-day period further positions in the same direction were added, these positions were to be liquidated along with the initial entry at the end of the five-day holding period. This was to prevent excessive idiosyncratic risk exposure on a single asset over a short period of time and to reset the algorithm in negative-spiralling scenarios, in which trend-confirming price movement continuously generates reversal signals and resulting in performance mimicking a Martingale strategy failure.

Results

The equity curve graph in Figure 1 demonstrates a positive performance over the five-year period with a negative performance during the second half of 2022, 2023 & 2024. This aligns with high-momentum and low volatility trend-driven market performance during these respective periods. Additionally, the heavy weighting of large-cap technology companies with high or increasing exposure to AI made the strategy more vulnerable to the strong positive trends in growth for AI stocks during this period.

Figure 2 demonstrates a positive net-returns of approximately 12%, with a below average Sharpe ratio of 0.8 and relatively better performing Sortino ratio of 1.06 with a limited max drawdown of 4%. The introduction of a 5% stop-loss and short holding period most likely contributed to the strategy's relatively better performance with risk management but also limited its ability to capture larger returns from more significant reversals.

Additionally, despite the positive performance, the S&P500 benchmark rose 40% during this period and in comparison, highlights the strategy's relatively mediocre performance.

Analysis

The strategy performed best during volatile market environments with a lack of long-term trends. This aligns with intuition as reversal signals are more likely to be false-positives when markets are driven by cyclical macro-economic fundamentals or transformative industry growth. However, a critical strength of the strategy is the utilisation of market neutrality, resulting in a more flexible strategy ambivalent to market-direction. Furthermore, the low correlation between indicators prevents excessive multicollinearity and ensure signals are orthogonal.

Despite this, Bollinger bands inherently rely on a soft assumption of stationarity of short-term volatility and makes the strategy vulnerable to regime changes in terms of trend of volatility. This can result in heavy drawdowns during periods of sustained high-momentum price movement, as the strategy lacks any indicator for testing changes to market conditions.

When comparing performance in Figure 1 to Figure 4 over this period, the strategy demonstrates a clear pattern of spikes in overperformance during major reversals including market drawdowns during February of 2025 and rally during October of 2023. However, anomalies also exist with significant portfolio drawdown from the second half of 2022 despite non-trending high market volatility, followed by overperformance from the beginning of 2023 despite the S&P500 breaking out into a positive low volatility trend.

This can be explained as due to the overexposure of the dataset to FANG stocks that relative to the market, experienced sharp rallies from previous drawdowns at the beginning of 2023. This volatility widened Bollinger bands and resulted in fewer false positives in the strategy, when prices approached the upper or lower bands.

Furthermore, despite a lack of cointegration and conclusive results for the pair-trading mean reversion strategy, the inability to reject the null hypothesis demonstrates the difficulty in finding stable cointegration relationships between individual equities. Theoretically, this strategy would perform much better when utilising derivative prices that have a stronger fundamental connection between pairs as opposed to equities, in which correlations are primarily from sharing a common industry or business model. These correlations can prove difficult to remain robust given idiosyncratic risk for each individual firm and occasionally inverse, as overperformance in an individual firm can reduce market share for its competitors in low-growth industries.

Conclusion

Overall, despite a net positive performance in the strategy over the five-year period, the strategy failed to reach the benchmark market performance. This can be mostly attributed to a lack of regime-switching mechanism in the strategy and inability to adapt to high-momentum environments.

A potential improvement could include incorporating dynamic momentum indicators on index ETFs to identify periods of regime switching. This could be combined with a Markov switching model to assign probabilistic weightings to the strength of the strategy's reversal indicators.

Appendix

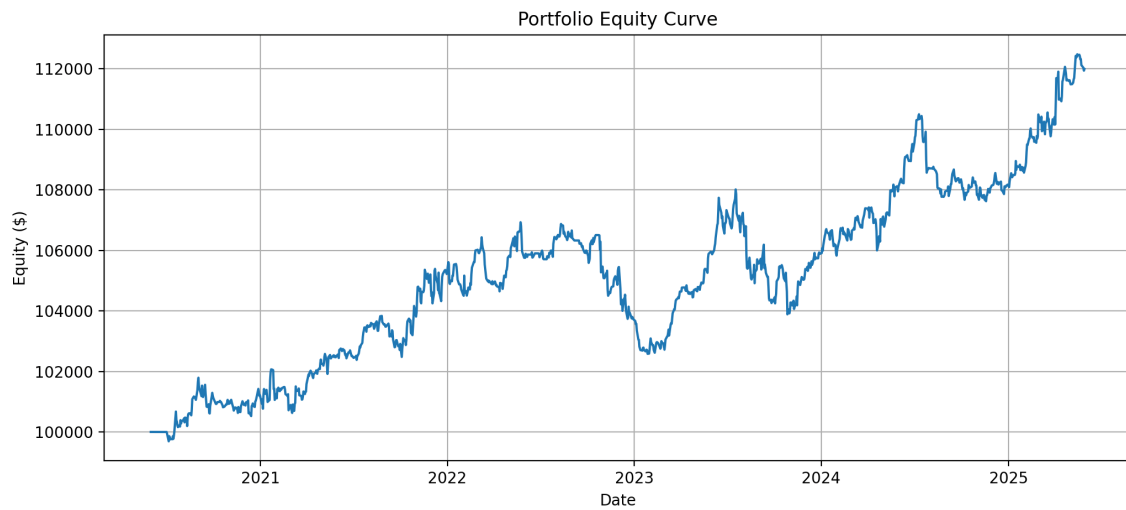


Figure 1

Total P&L	ROI	Sharpe Ratio	Sortino Ratio	Max Drawdown	Number of Trades	Win Rate	Average Profit (Closed Trades)
\$11,991.85	0.12	0.80	1.06	-0.04	553	0.48	\$19.85

Figure 2

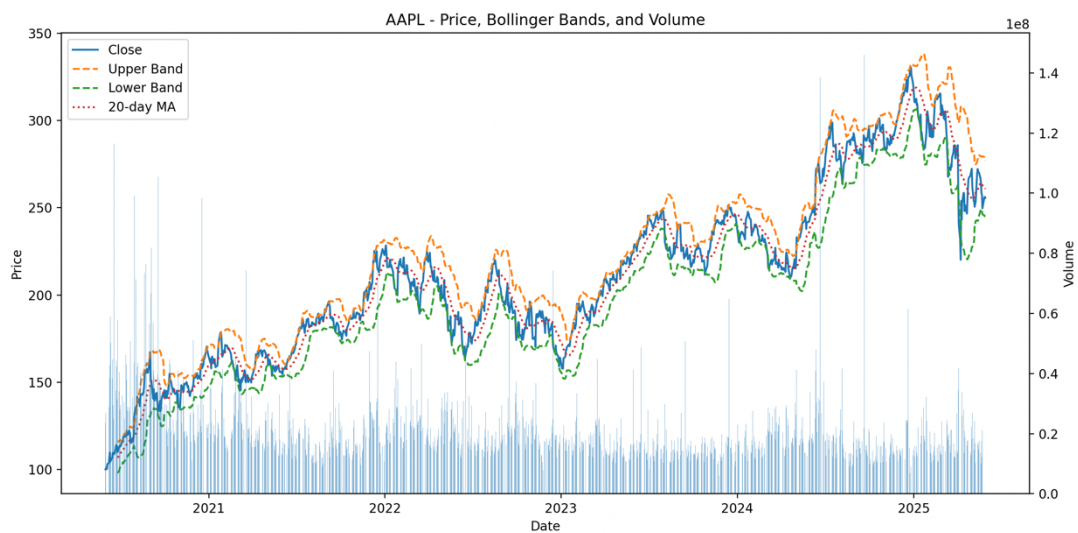


Figure 3 – Sample Ticker Indicators

S&P 500

INDEXSP: .INX

6,846.51

+3,183.05 (86.89%) ↑ past 5 years

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Figure 4 – S&P500 Performance